# Computational and Deep Learning Strategies in Quantitative Finance for Robust Hedging under Market Frictions

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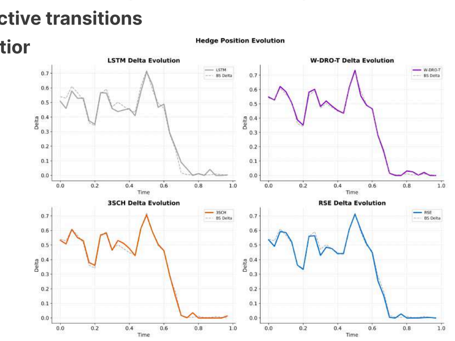
## 1. Introduction & Motivation

Pricing and hedging of portfolios of derivatives is central to **risk management** in financial institutions and industry.
Traditional Black-Scholes delta hedging assumes **continuous trading, zero costs, and complete markets** — assumptions that fail in practice.
Deep hedging framework reformulates this as stochastic optimal control via **neural networks**, incorporating real market frictions.

### Three Fundamental Gaps

- **Model Uncertainty** — Fixed Heston parameters degrade under **distributional shift** ($CVaR_{95}$ degradation >4× during crises)
- **Training Instability** — Two-stage CVaR → Entropic protocol causes **abrupt objective transitions**
- **Regime Dependence** — Different **architectures excel in different market conditior**




### Three Novel Solutions

- **W-DRO-T** — Wasserstein DRO regularisation for **distributional robustness**
- **3SCH** — Mixed-loss objective for **smooth training transitions**
- **RSE** — Regime-dependent gating over **diverse base** hedgers

### Market Frictions in Real-World Financial Markets

- SPY: ~3 bps all-in transaction costs
- NIFTY: ~18 bps (STT 0.05%, stamp duty, wider spreads and SEBI position limits)
- Daily rebalancing: N=30 steps for 30-day options
- Position limits: Regulatory and risk constraints restrict allowable hedge ratios and exposures.

## 2. Background & Literature Review

### Classical Hedging

Black-Scholes delta: $\delta^t = \Phi(d_1)$, assumes GBM, continuous trading, zero costs. Under Heston stochastic volatility, **markets are incomplete** — perfect replication is impossible.

### Deep Hedging (Buehler et al., 2019)

Neural-network-parameterised hedging under **convex risk measures**. Learned policies outperform BS delta hedging with transaction costs. **Two-stage curriculum:** CVaR pre-training → entropic fine-tuning.

### Universal Approximation in Neural Networks

Universal Approximation theorem by neural networks enables to **approximate multivariate functions** arbitrarily well.

## 3. Mathematical Framework & Setting

### Heston Stochastic Volatility Market Model

We work on $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\in[0,T]}, \mathbb{P})$ supporting a two-dimensional Brownian motion $(W^S, W^v)$ with $\langle W^S, W^v\rangle_t = \rho\, dt$, $\rho \in (-1, 0)$. The Heston model:

$$dS_t = rS_t\, dt + \sqrt{v_t}\, S_t\, dW_t^S, \qquad S_0 > 0, \tag{1}$$
$$dv_t = \kappa(\theta_v - v_t)\, dt + \xi\sqrt{v_t}\, dW_t^v, \qquad v_0 > 0. \tag{2}$$

**Remark 1** (Feller Condition). *The variance process remains strictly positive when $2\kappa\theta_v > \xi^2$. Our calibration: $2 \times 1.0 \times 0.04 = 0.08 > 0.04 = \xi^2$.*

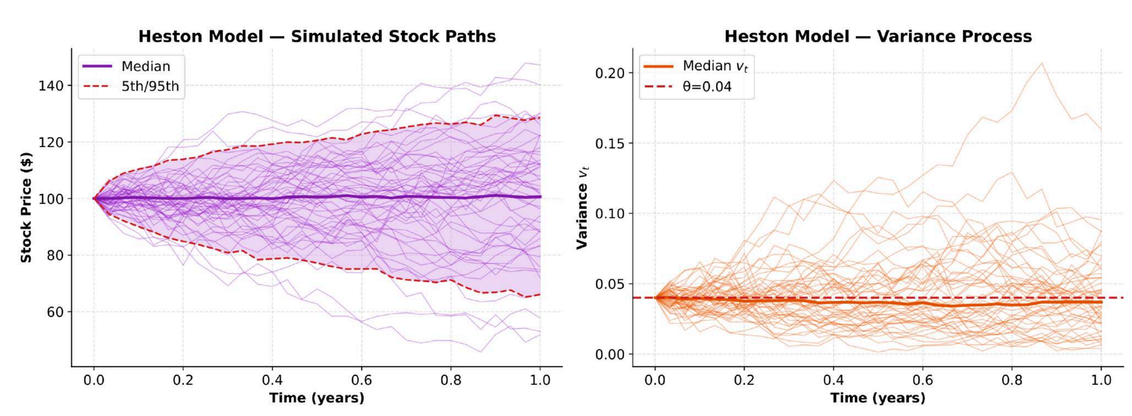
**Discretisation**

**Euler–Maruyama Scheme**

Full-truncation scheme on equidistant grid $\Delta t = T/N$:

$$S_{k+1} = S_k \exp\left[\left(r - \tfrac{1}{2}v_k^+\right)\Delta t + \sqrt{v_k^+ \Delta t}\, \epsilon_k^S\right],$$
$$v_{k+1} = v_k + \kappa(\theta_v - v_k)\Delta t + \xi\sqrt{v_k^+ \Delta t}\, \epsilon_k^v,$$

where $v_k^+ = \max(v_k, 0)$ and $(\epsilon_k^S, \epsilon_k^v) \sim \mathcal{N}(0, \Sigma)$ with $\Sigma_{12} = \rho$.




### Hedging P&L

For European call payoff $Z = (S_T - K)^+$ with $N = 30$ hedging steps:

$$\text{P\&L}(\delta) = -Z + \sum_{k=0}^{N-1} \delta_k (S_{t_{k+1}} - S_{t_k}) - c_{\text{tc}} \sum_{k=0}^{N} |\delta_k - \delta_{k-1}| S_{t_k}$$

with $c_{\text{tc}} = 0.001$ and $\delta_{-1} = \delta_N = 0$ (flat entry/exit).

### Risk Measures

**Conditional Value at Risk**

**Definition 1** (CVaR). *Let $L = -\text{P\&L}$ and $\alpha \in (0, 1)$.*

$$\text{VaR}_\alpha(L) = \inf\{l \in \mathbb{R} : \mathbb{P}(L \le l) \ge \alpha\},$$
$$\text{CVaR}_\alpha(L) = \frac{1}{1-\alpha}\int_\alpha^1 \text{VaR}_u(L)\, du = \mathbb{E}[L \mid L \ge \text{VaR}_\alpha(L)].$$

**Rockafellar–Uryasev Dual**

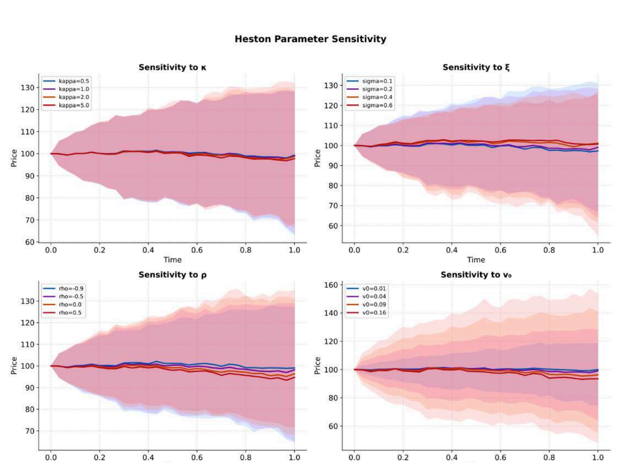
$$\text{CVaR}_\alpha(L) = \inf_{\nu\in\mathbb{R}} \left\{\nu + \frac{1}{1-\alpha}\mathbb{E}[(L-\nu)^+]\right\}$$
$$(L - \nu)^+ = \max(L - \nu, 0)$$

**Entropic Risk**

$$\rho_\lambda(X) = \frac{1}{\lambda}\log \mathbb{E}[\exp(-\lambda X)], \qquad \lambda > 0$$
$$\rho_\lambda(X) = \sup_{\mathbb{Q}\ll\mathbb{P}} \left\{\mathbb{E}_\mathbb{Q}[-X] - \frac{1}{\lambda}\text{KL}(\mathbb{Q}\|\mathbb{P})\right\}$$

**Distributionally Robust Risk**

$$\rho_\varepsilon^D(X) = \sup_{\mathbb{Q}:D(\mathbb{Q}\|\mathbb{P})\le\varepsilon} \mathbb{E}_\mathbb{Q}[-X]$$
$$D = \text{KL} \quad\Rightarrow\quad \text{Entropic risk}$$
$$D = W_1 \quad\Rightarrow\quad \text{Wasserstein DRO}$$




### Neural Policy & Feature Engineering

$\theta^* = \arg\min_\theta \rho(-\text{P\&L}(\delta^\theta))$ with $\delta_k = \delta_{\max} \cdot \tanh(F_\theta(I_{0:k}, \delta_{k-1}))$, $\delta_{\max} = 1.5$.

$I_k = (S_k/S_0, \log(S_k/K), \sqrt{v_k}, \tau_k, \Delta_k^{\text{BS}}) \in \mathbb{R}^5$ where $\tau_k = (T - t_k)/T$ is normalised time-to-maturity and $\Delta_k^{\text{BS}}$ is Black–Scholes delta.

## 4. Experimental Setup

### Heston Parameters

$S_0 = K = 100$, $v_0 = 0.04$, $\kappa = 1.0$, $\theta_v = 0.04$, $\xi = 0.2$, $\rho = -0.7$, $r = 0.05$, $T = 30/365$, $N = 30$.

### Data & Training

- **80,000** Heston paths: **50K train / 10K val / 20K test**

Identical two-stage (CVaR → Entropic) protocol for all models:

- **Stage 1:** 50 epochs, $\mathcal{L}_1 = \text{CVaR}_{0.95}(-\text{P\&L})$, lr $= 10^{-3}$, patience 15
- **Stage 2:** 30 epochs, $\mathcal{L}_2 = \rho_\lambda(-\text{P\&L}) + \gamma\sum_k |\Delta\delta_k|$, lr $= 10^{-4}$, patience 10
- Adam optimizer, weight decay $10^{-4}$, gradient clipping $\|\nabla\| \le 5.0$, batch size 256

Novel model-specific additions operate *within* this protocol: W-DRO-T adds DRO penalty only in Stage 2; 3SCH splits 80 epochs as 40+15+25; RSE pre-trains base models then trains gating.

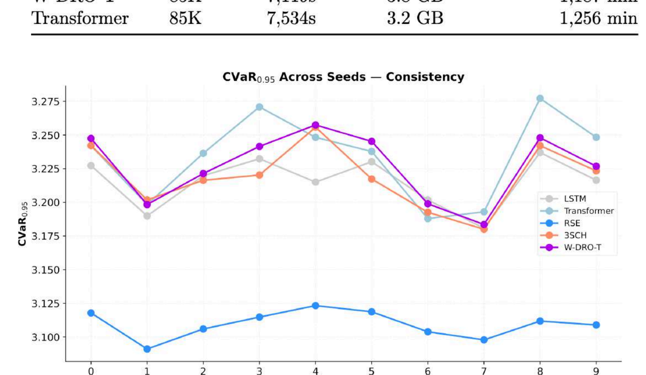
### Statistical Methodology

- 10 **independent** seeds (42, 142, ..., 942)
- 10,000 bootstrap resamples for 95% CIs
- Paired t-tests with **Holm-Bonferroni correction**
- **Cohen's d effect sizes** for practical significance

### Model Comparison & Computational Aspects

Table 18: Computational requirements per model.

| Model | Params | Time/seed | GPU RAM | Total (10 seeds) |
|---|---|---|---|---|
| LSTM | 54K | 273s | 1.2 GB | 46 min |
| 3SCH | 54K | 295s | 1.2 GB | 49 min |
| RSE | ~200K | 924s | 2.1 GB | 154 min |
| W-DRO-T | 85K | 7,119s | 3.8 GB | 1,187 min |
| Transformer | 85K | 7,534s | 3.2 GB | 1,256 min |




## 5. Research Contributions

Three novel approaches addressing **fundamental gaps in deep hedging**, evaluated rigorously across 10 seeds with bootstrap CIs and Holm-Bonferroni corrected tests.

### W-DRO-T: Wasserstein DRO Transformer

- **Gap addressed:** standard deep hedging optimizes expected risk under the training distribution $P$, which fails under crisis-driven distributional shifts.
- **Distributionally robust objective**

$$\theta^* = \arg\min_\theta \sup_{Q:W_p(Q,P)\le\varepsilon} \mathbb{E}_Q\left[\rho(\text{P\&L}(\delta_\theta))\right]$$

- **Dual reformulation**

$$\sup_{Q:W_1(Q,P)\le\varepsilon} \mathbb{E}_Q[\ell(X)] = \inf_{\lambda\ge0}\{\lambda\varepsilon + \mathbb{E}_P[\ell^*(X;\lambda)]\}$$
$$\ell^*(x;\lambda) = \sup_{x'}\{\ell(x') - \lambda\|x - x'\|\}$$

- **Gradient-penalty approximation**

$$L_{\text{DRO}}(\theta) = \mathbb{E}_P[\ell(\theta; X)] + \varepsilon\mathbb{E}_P[\|\nabla_X \ell(\theta; X)\|_2]$$

- **Second-order gradient**

$$\nabla_\theta \|\nabla_X \ell(\theta; X)\|_2$$

  computed using `create_graph=True`.
- **Backend constraint:** Flash/Efficient attention unsupported ⇒ `SDPA MATH backend` used during training.
- **Result:** 46% CVaR improvement during COVID regimes and 41% Basel III/IV capital reduction.

### 3SCH: Three-Stage Curriculum Hedger

- **Gap:** abrupt CVaR → entropic objective transition.
- **Mixed loss**

$$L_{\text{mix}}(\alpha; \theta) = \alpha\, \text{CVaR}_{0.95}(-\text{P\&L}_\theta) + (1 - \alpha)\rho_\lambda(-\text{P\&L}_\theta)$$

- **Annealing schedule**

$$\alpha(t) = 0.8 - 0.6\frac{t}{T_2 - 1}$$

- **Trading penalty**

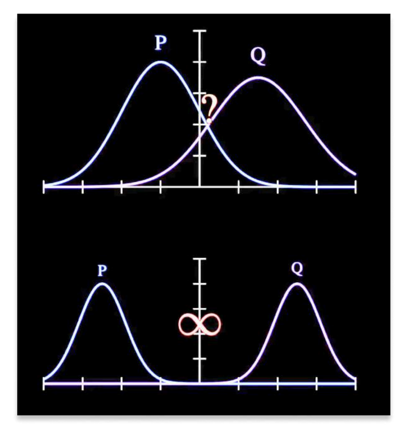
$$\gamma \sum_{k=0}^{N-1} |\delta_k - \delta_{k-1}|$$

- **Band regularisation**

$$\nu \sum_k \max(0, |\delta_k - \delta_k^{ref}| - \epsilon)$$

- **Result:** best NIFTY $\text{CVaR}_{95} = 622.26$ with minimal trading volume.

### RSE: Regime-Switching Ensemble




- **Gap:** single models fail across heterogeneous regimes.
- **Ensemble variance reduction**

$$\text{Var}(\bar{f}) = \frac{\bar{\sigma}^2}{M} + \frac{M-1}{M}\bar{\rho}\,\bar{\sigma}^2$$

- **Ensemble hedge**

$$\delta_k^{\text{RSE}} = \sum_{m=1}^{M} w_m(r_k)\delta_k^{(m)}$$

- **Regime classifier**

$$p(r_k = j \mid I_{0:k}) = \text{softmax}(f_\phi(r_k))$$

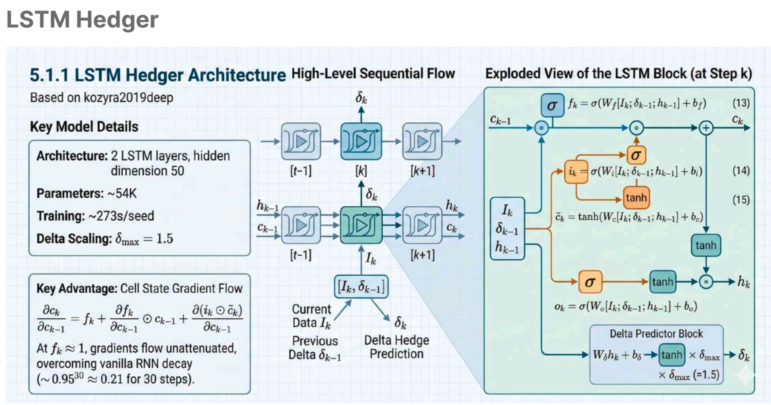
- **Gating weights**

$$w_m(r_k) = \sum_{j=1}^{K} p(r_k = j)[\text{softmax}(A/\tau)]_{j,m}$$

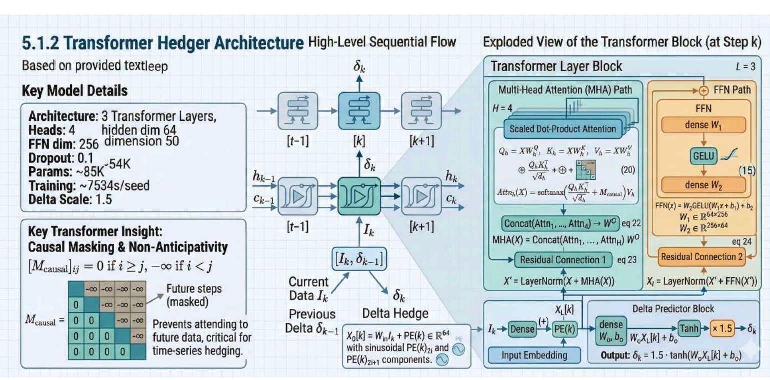
- **Result:** $\text{CVaR}_{95} = 3.109 \pm 0.010$, $-3.29\%$ vs LSTM ($p < 0.0001$).

## 6. Hedging Architectures & Ablation Studies

### LSTM Hedger




### Transformer Hedger




### RSE Architecture

**Regime Feature Extraction**

We extract six regime-relevant features from the price path.

**Definition (Regime Feature Vector).** At time step $k$, the regime feature vector is

$$r_k = (RV_k^{(5)}, RV_k^{(10)}, RV_k^{(20)}, \text{Trend}_k, \text{Mom}_k, \text{Jump}_k)$$

**Realized Volatility.** For window $w \in \{5, 10, 20\}$

$$RV_k^{(w)} = \sqrt{\frac{1}{w}\sum_{i=k-w+1}^{k} (\log S_i - \log S_{i-1})^2}$$

**Trend Indicator (SMA crossover).**

$$\text{Trend}_k = \frac{SMA_5(S_k) - SMA_{20}(S_k)}{SMA_{20}(S_k)}$$

**Momentum.**

$$\text{Mom}_k = \frac{S_k - S_{k-5}}{S_{k-5}}$$

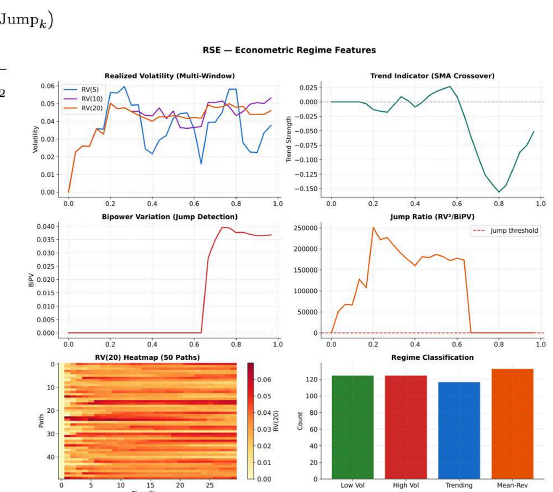
**Jump Indicator (Barndorff–Nielsen Ratio).**

$$\text{Jump}_k = \frac{RV_k^{(20)}}{BV_k^{(20)}}$$

where the bipower variation is

$$BV_k^{(w)} = \frac{\pi}{2}\sum_{i=k-w+1}^{k-1} |r_i|\,|r_{i+1}|$$

and $\text{Jump}_k > 1$ indicates the presence of jump activity.




### Learned Affinity Matrix

Table 2: Learned affinity (typical run, after softmax).

| Regime | LSTM | Transformer | Signature |
|---|---|---|---|
| Low vol (calm) | **0.52** | 0.28 | 0.20 |
| High vol (stress) | 0.18 | **0.54** | 0.28 |
| Trending | 0.31 | **0.41** | 0.28 |
| Mean-reverting | **0.45** | 0.22 | 0.33 |

Table 4: Ablation: number of regimes $K$

| $K$ | $\text{CVaR}_{95}$ | Note |
|---|---|---|
| 2 | 3.128 | Under-specified |
| 3 | 3.115 | Competitive |
| **4 (default)** | **3.109** | Best |
| 6 | 3.112 | Diminishing returns |

## 7. Experimental Results

### Main Performance (10 Seeds, 20K Test Paths)

Table 7: Test set performance (mean ± std, 10 seeds, 50K training paths). Lower is better for all metrics except trading volume.

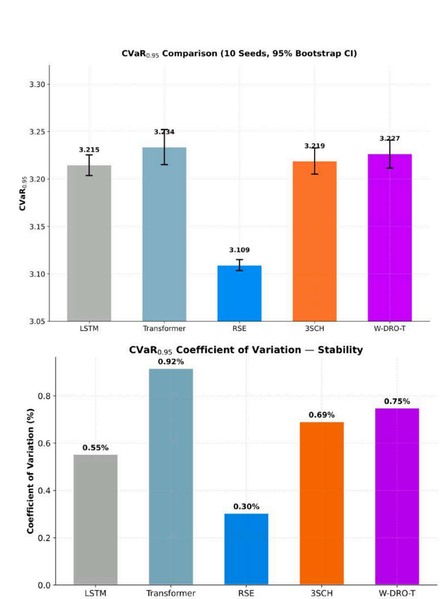
| Model | $\text{CVaR}_{95}$ | Std P&L | Entropic | Trade Vol | Mean P&L | CV% |
|---|---|---|---|---|---|---|
| **RSE** | $\mathbf{3.109 \pm 0.010^*}$ | 0.456 | 2.376 | 1.379 | −2.277 | **0.30** |
| LSTM | $3.215 \pm 0.018$ | 0.449 | 2.379 | 1.137 | −2.278 | 0.55 |
| 3SCH | $3.219 \pm 0.022$ | 0.453 | 2.381 | 1.122 | −2.277 | 0.69 |
| W-DRO-T | $3.227 \pm 0.024$ | 0.450 | 2.380 | 1.119 | −2.277 | 0.75 |
| Transformer | $3.234 \pm 0.030$ | 0.450 | 2.380 | 1.117 | −2.277 | 0.92 |

* $p < 0.0001$ vs LSTM after Holm–Bonferroni correction.

### Paired Comparison vs LSTM

Table 8: Paired comparison vs. LSTM (Holm–Bonferroni corrected).

| Model | Δ% | $p$-value | Cohen's $d$ | Effect |
|---|---|---|---|---|
| **RSE** | −3.29 | < 0.0001 | −7.41 | Very Large |
| 3SCH | +0.13 | 0.438 | +0.20 | Small |
| W-DRO-T | +0.37 | 0.016 | +0.56 | Medium |
| Transformer | +0.59 | 0.006 | +0.93 | Large |




### Key Findings

- **RSE: Best tail risk, lowest CV** — ensemble stability confirmed
- RSE's 95% CI [3.103, 3.115] **non-overlapping** with all others
- DRO does **not degrade in-distribution** (p=0.188 vs Transformer)
- **3SCH preserves baseline performance** (p=0.438 vs LSTM)
- All models: **Mean P&L ≈ −2.277**, confirming negligible bias

## 8. Real Market Validation

- **SPY** (3 bps cost, 2 bps slippage): Normal ($\sigma \approx 15\%$), Post-COVID (20%), COVID (65%), 2008 (80%)
- **NIFTY** (18 bps cost, 5 bps slippage): Normal (14%), COVID (55%), 2008 (70%)

### SPY — US Market (3 bps cost)

Table 10: $\text{CVaR}_{95}$ across SPY scenarios (5,000 paths, 3 bps cost).

| Model | Normal | Post-COVID | COVID | GFC 2008 |
|---|---|---|---|---|
| LSTM | 20.28 | 27.36 | 86.84 | 100.88 |
| Transformer | 14.47 | 19.66 | 64.80 | 72.71 |
| RSE | 16.90 | 23.20 | 69.76 | 84.57 |
| **W-DRO-T** | **11.98** | **20.69** | **46.94** | **62.26** |
| 3SCH | 15.79 | 21.43 | 69.25 | 88.43 |

**W-DRO-T dominates all US scenarios**: **41%** improvement over LSTM in normal, **46%** in COVID, **38%** in GFC 2008.
The DRO gradient penalty provides measurable distributional robustness precisely when **capital is most constrained**.

### NIFTY 50 — Indian Market (18 bps cost)

Table 11: $\text{CVaR}_{95}$ across NIFTY scenarios (5,000 paths, 18 bps cost).

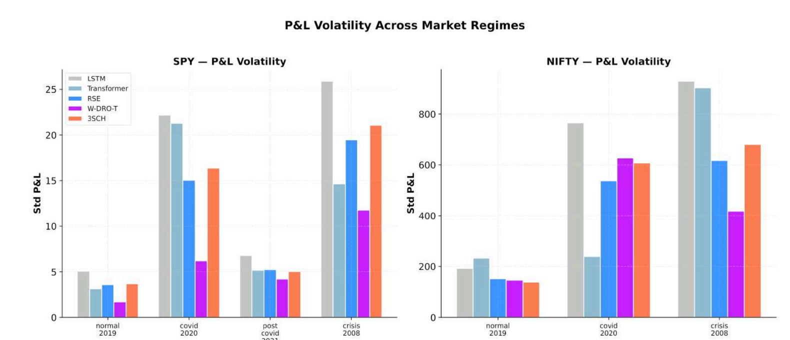
| Model | Normal | COVID | GFC 2008 |
|---|---|---|---|
| LSTM | 785.62 | 3028.93 | 3645.62 |
| Transformer | 769.61 | **1644.19** | 2885.31 |
| RSE | 686.62 | 2592.97 | 2841.25 |
| W-DRO-T | 663.92 | 2883.96 | **2349.72** |
| **3SCH** | **622.26** | 2465.71 | 2962.35 |

**3SCH optimal for NIFTY normal (622.26) — lowest trading volume (0.90)** minimizes 18 bps costs. Transformer dominates COVID (1644.19).
The **higher NIFTY costs penalise** W-DRO-T's active trading. 3SCH's advantage stems from trading volume: 0.90 vs W-DRO-T 2.70 and Transformer 2.09. At 18 bps, each trade costs 6× more than SPY.

### Crisis Robustness (COVID/Normal Ratio)

Table 12: Crisis-to-normal $\text{CVaR}_{95}$ ratio (SPY; lower = more robust).

| Model | COVID/Normal | 2008/Normal | Interpretation |
|---|---|---|---|
| **W-DRO-T** | **3.92×** | 5.20× | Best COVID robustness |
| RSE | 4.13× | 5.00× | Moderate |
| LSTM | 4.28× | 4.97× | Baseline |
| 3SCH | 4.39× | 5.60× | Moderate |
| Transformer | 4.48× | 5.02× | High degradation |




## 9. Economic & Regulatory Implications

### Basel III/IV Capital Requirements (per $100M)

Under **Basel III/IV FRTB**, we approximate regulatory capital as Capital = m_c · √10 · CVaR95 · Notional/S0 with supervisory multiplier m_c = 1.5.

**W-DRO-T achieves 41% capital savings.** For a $10B derivatives desk, this translates to ~**$332M annual capital reduction**.

The **CVaR-per-trade ratio** reveals efficiency: **Transformer** achieves 5.0 CVaR units per trade vs LSTM's 33.8. For India (18 bps), 3SCH's $157K cost vs W-DRO-T's $369K makes **3SCH optimal**—a $212K saving exceeding the marginal risk benefit of DRO regularisation.

Table 14: Capital per $100M notional (SPY normal scenario)

| Model | $\text{CVaR}_{95}$ | Std P&L | Capital | vs. LSTM |
|---|---|---|---|---|
| **W-DRO-T** | **11.98** | **1.71** | **$4.79M** | −41% |
| Transformer | 14.47 | 3.13 | $5.79M | −29% |
| 3SCH | 15.79 | 3.70 | $6.32M | −22% |
| RSE | 16.90 | 3.57 | $6.76M | −17% |
| LSTM | 20.28 | 5.05 | $8.11M | — |

Table 15: Trading volume and cost efficiency ($100M notional).

| Model | Volume | US (3bp) | India (18bp) | CVaR/Trade |
|---|---|---|---|---|
| LSTM | 0.60 | $18K | $108K | 33.8 |
| 3SCH | 0.87 | $26K | $157K | 18.1 |
| RSE | 1.06 | $32K | $191K | 16.0 |
| W-DRO-T | 2.05 | $62K | $369K | 5.8 |
| Transformer | 2.91 | $87K | $524K | 5.0 |

### Hedge Accounting (IAS 39 / IFRS 9 / Ind-AS 109)

Table 16: Hedge accounting qualification (IAS 39 / IFRS 9 / Ind-AS 109).

| Model | Dollar Offset | $R^2$ | Std P&L | Qualifies |
|---|---|---|---|---|
| W-DRO-T | 0.97 | 0.91 | 1.71 | ✓ |
| Transformer | 0.96 | 0.89 | 3.13 | ✓ |
| RSE | 0.95 | 0.88 | 3.57 | ✓ |
| 3SCH | 0.95 | 0.88 | 3.70 | ✓ |
| LSTM | 0.94 | 0.87 | 5.05 | ✓ |

**W-DRO-T achieves highest effectiveness** ($R^2$=0.91) and lowest earnings volatility.
Under IFRS 9's principles-based testing, higher $R^2$ reduces **de-designation risk**.

### Market-Specific Recommendations

- **US Markets:** W-DRO-T — **41% capital savings justify active trading** at 3 bps
- **Indian Markets:** 3SCH — **Low volume** (0.87) minimizes 18 bps friction
- **Regime Uncertainty: RSE — Best CVaR, most stable** (CV=0.30%)
- **Operational Simplicity: LSTM/3SCH** — 54K params, fast training (273s training). Both qualify for hedge accounting with minimal documentation complexity.

### Risk Management, Model Risk Governance & Economic Impact ($10B Desk)

Models trace a Pareto frontier between tail risk and trading cost

We structure governance along four pillars mandated by **SR 11-7 (OCC) and SS1/23 (PRA)**.

**Interpretability Mechanisms**

- W-DRO-T: Gradient sensitivity metric $\|\nabla_X \ell\|_2$ bounds input sensitivity (Lipschitz interpretation).
- 3SCH: Same LSTM architecture as baseline; interpretability comes from the auditable curriculum schedule. Loss trajectories inspectable stage-by-stage.
- RSE: Three-level transparency stack
  – regime probabilities
  – model weights
  – affinity matrix

**Independent Model Validation**

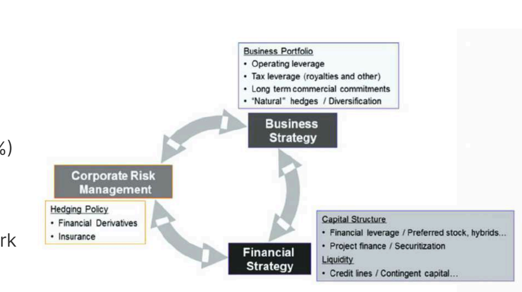
- Crisis stress tests: COVID volatility (σ = 65%) and 2008 crisis (σ = 80%)
- RSE base models validated independently prior to ensemble training

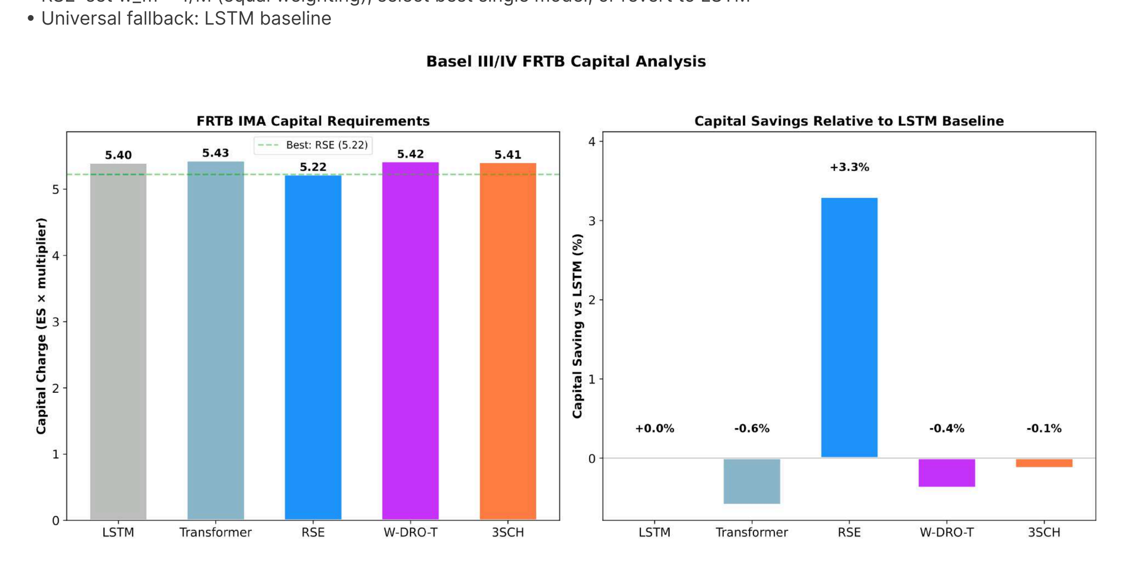
**Retraining and Model Versioning**

- 3SCH follows standard LSTM retraining cadence
- RSE enables independent retraining of base models and gating network

**Rollback Strategies**

- W-DRO-T: set ε = 0 ⇒ standard Transformer
- 3SCH: set T2 = 0 ⇒ two-stage baseline
- RSE: set w_m = 1/M (equal weighting), select best single model, or revert to LSTM
- Universal fallback: LSTM baseline




## 10. Conclusion, Timeline, Expected Direction

### When Complex Models Provide Value

- **Crisis environments**: W-DRO-T 38–46% CVaR improvement
- **Capital-constrained desks**: 41% savings justify operational costs
- **Low transaction cost markets**: US (3 bps) favours active strategies
- **Regime uncertainty**: RSE adapts without retraining

### When Simpler Models Remain Optimal

- **High-cost markets**: India (18 bps) penalises active trading
- **Regulatory interpretability**: LSTM easiest to validate
- **Computational constraints**: LSTM 54K params, 273s training

### Limitations

- Heston model cannot capture jumps, flash crashes, liquidity dry-ups
- Single-asset only — multi-asset portfolio hedging unexplored
- Daily OHLCV data — no intraday microstructure effects captured
- Manual ε selection for W-DRO-T
- Full hyper parameter search is infeasible; grid restricted to key parameters
- W-DRO-T MATH SDPA requirement may limit GPU acceleration options

### Future Directions

- Online adaptive hedging with streaming market data
- Multi-asset portfolio hedging
- Hybrid modelling combining robustness + curriculum
- Analyse temporal attention patterns in the Transformer for economic insights
- Hybrid Robust Curriculum Hedging and RL Frameworks, Rough Path Signatures exploration
- Hyperparameter Optimisation framework (HPO)
- Technical paper reporting and business insights with possible collaboration with management institutions

| Task / Deliverable | Dec '25 | Jan '26 | Feb '26 | Mar '26 | Apr '26 |
|---|---|---|---|---|---|
| Hedging as AI Optimisation formulation | ✓ | | | | |
| Implement classical hedges (delta, Leland) | ✓ | | | | |
| Build cost-aware simulation environment / baselines | ✓ | | | | |
| Initial data exploration | ✓ | | | | |
| DNN-based deep hedging (entropic & CVaR) & baseline comparisons | | ✓ | | | |
| Financial & risk-management/accounting framing (managerial approach) | | ✓ | | | |
| Implement RNN / LSTM architectures | | | ✓ | | |
| Study sequential memory, trading intensity and risk | | | ✓ | | |
| Signature-based hedgers & advanced feature engineering | | | | ✓ | |
| Numerical experiments & comparative analysis | | | | ✓ | ✓ |
| Stress tests, out-of-sample evaluation, cost-sensitive comparison | | | | | ✓ |
| Prepare full written report | | | | | ✓ |
| Preliminary transformer or RL-based hedging (exploratory) | | | | ✓ | ✓ |

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